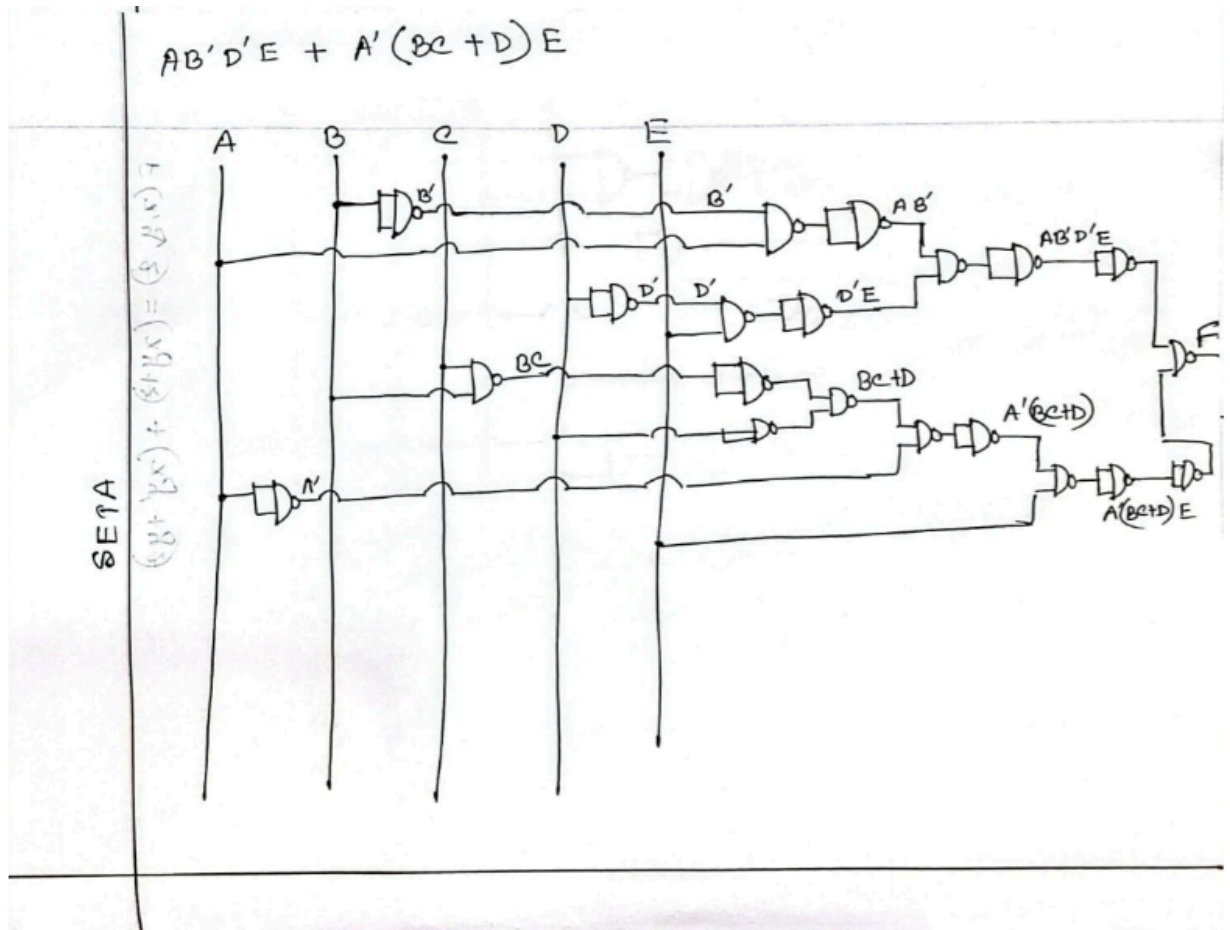


SET A:

1. Draw the following functions using NAND gates only: [7.5 Marks]

$$1. AB'D'E + A'(BC + D)E$$

NB: You can't simplify the above functions and then draw using NAND gates. You have to draw based on the function given in question.



2. Convert the following function to its corresponding canonical **SOP** form using the distributive laws. Also, write down the **Minterms** of the function. [7.5 Marks]

$$F(a,b,c,d) = a + b'(c + a')$$

$$\begin{aligned} &= a + b'c + a'b' \\ &= a(b + b') + (a + a')b'c + a'b'(c + c') \\ &= ab(c + c') + ab'(c + c') + ab'c + a'b'c + a'b'c + a'b'c' \\ &= abc + abc' + ab'c + ab'c' + ab'c + a'b'c + a'b'c' \\ &= abc(d + d') + abc'(d + d') + ab'c(d + d') + ab'c'(d + d') + a'b'c(d + d') + a'b'c'(d + d') \\ &= abcd + abcd' + abc'd + abc'd' + ab'cd + ab'cd' + ab'cd + ab'cd' + a'b'cd + a'b'cd' + a'b'cd + a'b'cd' \\ &+ a'b'c'd' = \sum m(0, 1, 2, 3, 8, 9, 10, 11, 12, 13, 14, 15) \end{aligned}$$

SET B:

1. Draw the following functions using NAND gates only: [7.5 Marks]

$$A(B'D + C)'E + A'BDE'$$

NB: You can't simplify the above functions and then draw using NAND gates. You have to draw based on the function given in question.

